Machine Learning Engineer

Assessment Instructions:

- Objective
  - Use data/train_test.csv as the labeled development data.
  - Decide how to split and validate the development data.
  - Explore, clean, engineer features, validate, and model the data as you see fit.
  - Run your trained model on every load in data/validation.csv.
    - Data/validation.csv contains the 12,000 loads requiring final predictions. Each row contains the load features and a unique load_id.
    - Run your trained model on every row in data/validation.csv.
    - data/validation_predictions_template.csv contains the same 12,000 load_id values.
    - Fill the template's predicted_rate column and save the completed file as validation_predictions.csv.
  - Please refer to the README.md for setup instructions.

- Submit
  - An accessible GitHub repository containing your solution code, dependencies, and run instructions.

  - validation_predictions.csv containing exactly:

    - load_id,predicted_rate

  - A PDF or DOCX report containing:

    - Your approach to validate test/train data and split.
    - The fixed December prediction chart produced by the provided score.py

  - A 2-3 minute loom going over the following:
    - Key findings from exploring the data
    - Any data-quality issues identified and how you addressed them
    - Reasoning behind the chosen model
    - Training and validation approach, including how the data was split
    - A brief walkthrough of the most important parts of your code